THE LAST MILE

The last mile of a diabetes result.

The lab calls the doctor. Nobody calls the patient.

A patient-closure ledger for abnormal and critical diabetes results. A value echo, a documented contact, an unreachable disposition, a next step, and a receipt.

IIT Bombay Health-a-thon 2026
Diabetes track · Build Sprint

The lab is done when it delivers. The clinic is done when the patient is reached.

CLOSURE RECEIPT

Asha Rao · P0482

Potassium 5.9 mmol/L · ACE inhibitor start

03:10	**Lab releases the report**	Critical, lab flagged
03:12	**Clinician echoes 5.9**	Lab notifier + read-back
03:40	**Patient contacted in Hindi**	Call · caregiver confirmed
03:55	**Next step: repeat potassium**	Teach-back recorded

CLOSURE CERTIFICATE

Chain intact · evidence present · GLF-F656915ED9

Unreachable is a disposition, not a false informed.

THE RESULT STOPS AT THE DOCTOR

Only one in four critical values reached the treating clinician.

25.8%

critical values reached a clinician

52.9%

went to nursing staff

90%

of never-notified values were outpatients

A 2015 audit of 1,187 critical values at an Indian tertiary hospital found a written policy, a log book and a read-back rule. The failure was not the rule. It was the last mile: the result reached a desk, not the person who could act. The same paper found 9 in 10 never-notified values were outpatients.

LAB
releases the report

→

DOCTOR
echoes the value

→

PATIENT
is contacted, or not

14% of positive results never returned (NMJI 2015). 6.8% to 62% of abnormal ambulatory results were not followed up (J Gen Intern Med 2012).

CLOSE THE LAST MILE, NOT ANOTHER INBOX

Four controls. One honest closure.

The lab already notifies the doctor. We make the patient side visible, and we do not let a click stand in for a conversation.

01 VALUE ECHO

The clinician types the printed value and selects the printed flag. For a critical result, the lab caller and the value read back are recorded.

02 PATIENT CONTACT

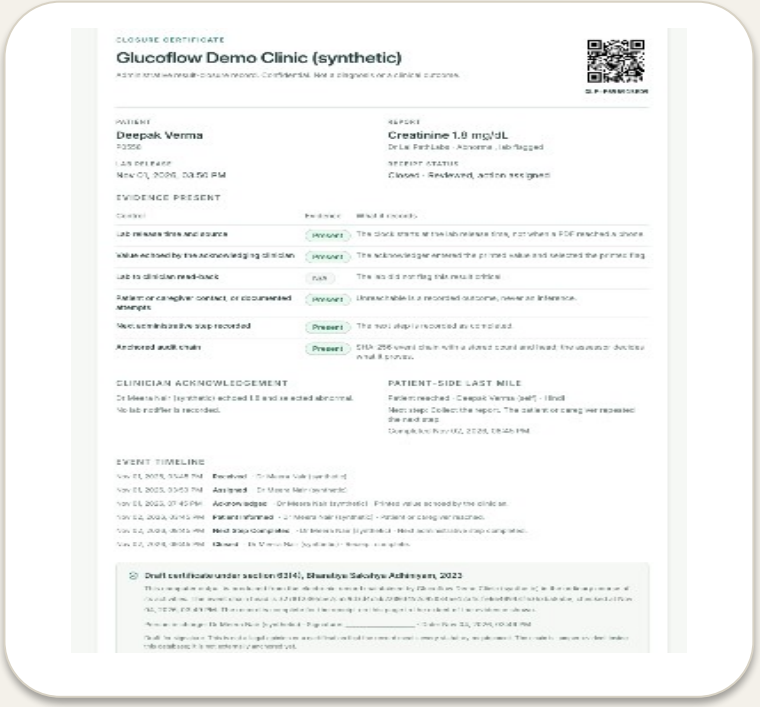
Patient or caregiver, channel, language, who was spoken to. Every failed attempt is a row, not a silent assumption.

03 AN HONEST DISPOSITION

Informed, declined, unreachable, invalid number, deceased, transferred. Two attempts before an unreachable close. Never a false informed.

04 NEXT STEP + PROOF

Appointment, repeat test, specialist, collect report. Completion is tracked. The certificate maps evidence, not a compliance verdict.


CLOSURE CERTIFICATE

Glucoflow Demo Clinic (synthetic)

PATIENT
Deepak Verma

REPORT
Creatinine 1.8 mg/dL

EVIDENCE PRESENT

CLINICIAN ACKNOWLEDGEMENT

PATIENT-SIDE LAST MILE

EVENT TIMELINE

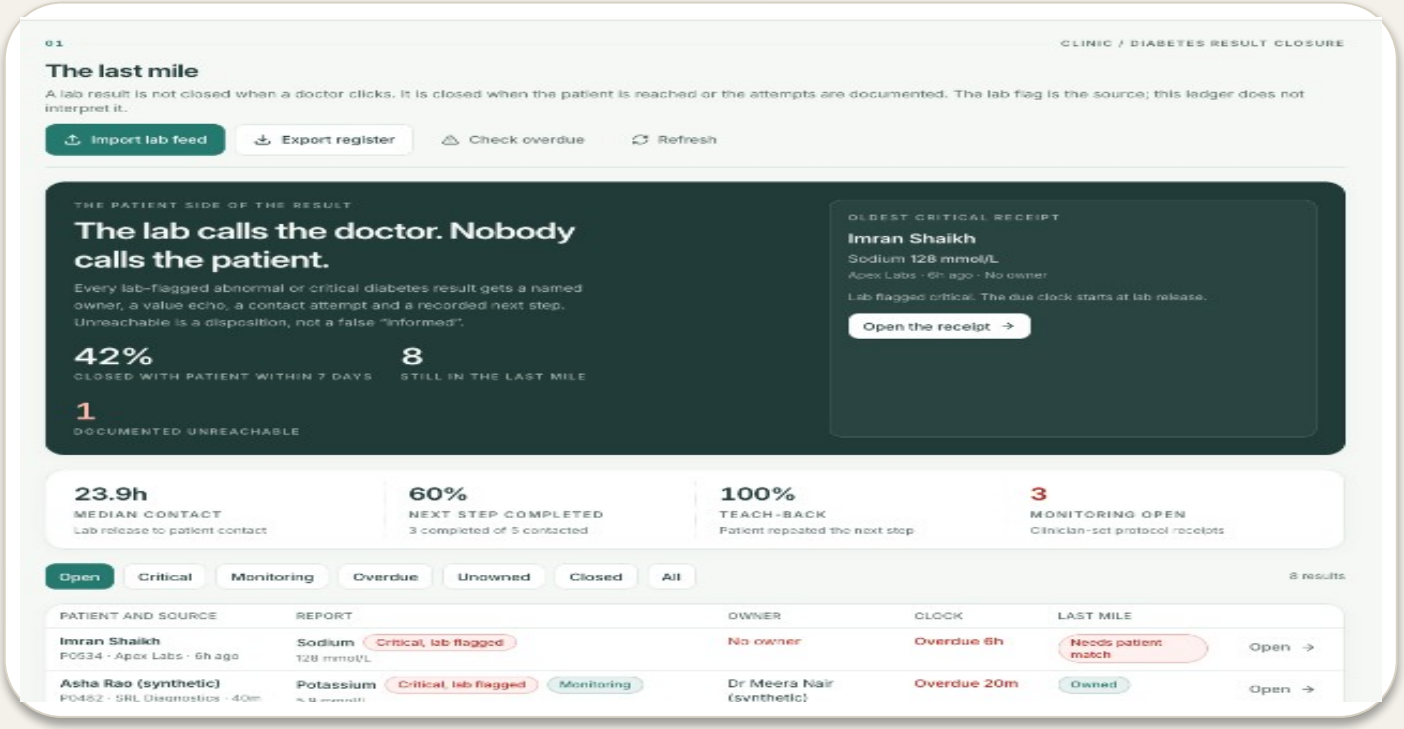
Draft certificate under section 63(4), Bharatiya Sakshya Adhiniyam, 2023

Closure certificate: evidence, timeline, section 63(4) draft.

THE LOOP RUNS TODAY

A working closure ledger.

Twelve synthetic diabetes receipts across nine patients and four labs. Five clinician-set monitoring protocols. The demo imports a lab feed and holds an unmatched report.



LAB FEED

CSV import; unmatched hold

IDENTITY

name + clinic ID candidates

VALUE ECHO

printed value and flag

LAST MILE

attempts + unreachable close

ESCALATION

timer + queued notification

PROOF

anchored chain + public stage

No live LIS or WhatsApp. The production adapter is the next gate.

THE RESULT A DIABETES DRUG MAKES NECESSARY

Start with the potassium and eGFR after a new ACE inhibitor.

The clinician sets the protocol. The lab flag sets criticality. The product does not infer a screening deadline or a clinical action.

01

START OR TITRATE

ACE inhibitor or ARB

The prescriber sets the monitoring protocol.

02

2 TO 4 WEEKS

Potassium + eGFR

The due window is a protocol, not an inference.

03

RESULT ARRIVES

Lab flag copied

Critical, abnormal or normal. No interpretation.

04

CLOSE THE LOOP

Value echo + contact

Patient reached, or attempts documented.

05

NEXT STEP

Repeat, review, attend

Tracked to completion; certificate ready.

The same receipt covers SGLT2i, MRA, annual uACR/eGFR, a critical potassium, or an abnormal HbA1c. 101 million people in India live with diabetes; the first market is the lab that already sends the report.

MEASURED, AND WHAT IS NOT

The build is measured. The demand is not.

WHAT WE RAN

188	unit and integration tests, 31 files
12 / 12	result and landing browser tests, desktop + mobile
0	lint findings on ledger, drawer, certificate, slip and verifier
12	synthetic receipts, 9 patients, 4 labs
4	new integration tests for value echo, unreachable close, chain and matching

WHAT WE HAVE NOT VALIDATED

— No buyer interview, baseline audit, paid pilot or LOI.

— No live LIS, HL7/FHIR or WhatsApp send.

— No consent capture, DPA or retention runbook.

— No external anchor: the chain is tamper-evident inside one database.

— No clinical validation. No diagnosis, triage or treatment advice.

We will not claim these.

The next number is a baseline audit, not a TAM.

SELL THE CLOSURE RATE, NOT ANOTHER SEAT

One lab chain. Many clinics. A measurable last mile.

THE BUYER

Lab chain quality head, or a hospital group quality cell.

The lab already sends the report. It pays for proof that the last mile closed and for a differentiator to referring doctors. The clinic uses the inbox free; the diabetes clinician is the champion.

PILOT

₹60,000 / 90 days. One lab, one diabetes OPD, one ward. 100 de-identified reports for the baseline.

THE METRIC

Closure rate within 7 days.

1. Abnormal or critical result with a named owner
2. Value echo + lab read-back where critical
3. Patient contacted, or attempts documented
4. Next step recorded; completion tracked

The network is the expansion, not the product.

No TAM claim. The first proof is one lab, one cohort, one signed continuation.

THREE SITES. ONE MEASURED CLOSURE RATE

A baseline audit, not a demo.

100 consecutive abnormal or critical diabetes reports per site. Eight weeks. A baseline audit, then the closure workflow, then a paid continuation. If the first two gates fail, the thesis stops.

01 **ONE LAB OR OPD SPONSOR**
A quality head who will share 100 de-identified reports.

02 **ONE CLINICAL LEAD**
A practising diabetes clinician who signs the protocol and the pilot.

03 **ONE HONEST NUMBER**
Closure rate within 7 days, unreachable documented, zero false informed.

THE CLOSING LINE

The lab is done when it delivers. The clinic is done when the patient is reached.

The result nobody told.

Synthetic demo. No real patient data. No diagnosis or treatment advice.